

Atishaya Maharjan

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EDUCATION & HONOURS

BSc. Joint Honours: Computer Science & Mathematics, Minor in Statistics Aug. 2022 – Present
University of Manitoba *Winnipeg, Canada*

International Undergraduate Student Scholarship | Dean's Honour Roll

RESEARCH EXPERIENCE

Undergraduate Researcher – Algorithms & Computational Geometry Mar. 2024 – Present
Geometric, Approximation, and Distributed Algorithms (GADA) Lab *Winnipeg, Canada*

- Optimizing collaborative drone-delivery routes for logistics by developing exact and approximation algorithms with dynamic programming, exchange-point geometry, and structural analysis.
- Developed an approximation algorithm for Hamiltonian-path optimization in spatially embedded graphs [↗](#), with visualization and randomized testing infrastructure covering **100+** synthetic instances.
- Improved the performance of a noncrossing spanning-tree research library by **30%** and introduced stronger **Python** testing, documentation, and development practices.

Undergraduate Researcher – Combinatorics & Graph Theory Jan. 2024 – Jul. 2025
Department of Mathematics, University of Manitoba *Winnipeg, Canada*

- Established a coefficientwise inequality underlying the Hurlbert–Kamat property for perfect m -ary trees [↗](#) and derived a closed-form expression for their independence number.
- Applied generating functions, polynomial recurrences, induction, and coefficientwise analysis to develop the mathematical argument.
- Implemented **SageMath** and **Python** experiments across **1,000+** instances to test conjectures, verify symbolic identities, and validate intermediate results.

Machine Learning Researcher May 2025 – Present
Terrabyte Machine Learning Research Group *Winnipeg, Canada*

- Extended diffusion-based learning to higher-order relational data by implementing **4** Hypergraph Neural Diffusion variants using **PyTorch** and **PyTorch Geometric**.
- Built a reproducible **MLflow** pipeline tracking **25+** experiments across **5+** datasets.
- Tracked configurations, metrics, and evaluation results to support reproducible model comparison and analysis.

PROFESSIONAL EXPERIENCE

Software Engineer Mar. 2024 – Present
LG Electronics *Englewood Cliffs, USA (Remote)*

- Shipped **Next.js/Redux/TypeScript** B2B platform features for **1,000+** vendors, contributing to a **25%** user-engagement lift.
- Designed analytics and tagging frameworks adopted by **6+** product teams, enabling real-time measurement and data-driven feature prioritization.
- Built AI-powered global search with **35 ms** lower latency and checkout workflows that drove a **15%** conversion lift across promo-code and magic-link flows.

Software Engineer | Vibecore – *Cloud Infrastructure & CI/CD*

- Automated CI/CD deployments from **GitHub Actions** to **AWS EC2**, standardizing releases for EC2-hosted applications.
- Configured **S3**, **DynamoDB**, and **IAM** permissions for cloud-connected application workflows.
- Provisioned and maintained **EBS/EFS** storage for application data and deployment infrastructure.

Software Engineer Jun. 2025 – Jan. 2026
Hoek Insights *Amsterdam, Netherlands (Remote)*

- Engineered a **Python** package for **Three.js** visualizations in **Jupyter**, reducing render time by **38%** and bundle size by **25%** through **React** renderer optimization.
- Shipped a **FastAPI/PostgreSQL** MVP with **18+** endpoints and **10** product workflows.
- Maintained below **200 ms** median staging latency and **95%** integration-test coverage through automated API testing.

Software Engineer

Aug. 2019 – Apr. 2024

Timeero

Phoenix, USA (Remote)

- Led frontend development of a **TypeScript/React/Vite/Mapbox** mileage-tracking application supporting **1,000+** active users.
- Shipped **15+** geospatial workflows for segment mapping, mileage capture, and field operations.
- Improved time-capture accuracy by **30%** through Mapbox-based mapping and production workflow refinements.

SELECTED RESEARCH AND TECHNICAL PROJECTS

Brainstorm | *Authorization-Aware RAG, Data Systems*

- Built a role-aware knowledge-management platform integrating **Jira**, **PostgreSQL**, **Redis**, and **Pinecone** through retrieval-augmented generation.
- Implemented source grounding and role-based access control across connected organizational data systems.
- Built multi-model routing with fallback policies balancing latency, reliability, and response quality.

WinLocate | *Applied Data Engineering, Mapping*

- Integrated **Socrata** open data into an applied data platform for city and relocation analysis.
- Built city comparisons, interactive zoning maps, budget tools, and relocation decision support workflows.
- Won **1st place** among **15** teams at the Winnipeg Open Data Datathon 2026.

Get Good Grades | *Electron, React, TypeScript, YAML*

- Built a decentralized flashcard viewer that ingests **.yaml** flashcard files from anywhere in the file system.
- Gave users full control over flashcard content and scheduling logic instead of relying on centralized storage.
- Presented the project at DevHacks 2026.

RESEARCH AWARDS

S.U.R.E. Poster Competition – Third Place, Computational Science Category

Aug. 2024

- Received third place for the research poster “*Minimizing Maximum Interference in 1-Dimensional Networks*” at the University of Manitoba.

CSSA Design & Craft Case Competition – 1st Place

2026

AccelR8 Prairie AI Hackathon – Winner

2026

Winnipeg Open Data Datathon – 1st Place among 15 teams

2026

CSSA Capture the Flag – 1st of 41 teams (2026), 2nd of 25+ teams (2025)

.devHacks – Most Chaotic Evil Hack among 20+ teams

TECHNICAL SKILLS

Programming: Python, C++, C, TypeScript, JavaScript, Java, R, SQL, Bash, SageMath

Scientific Computing & ML: PyTorch, PyTorch Geometric, NumPy, SciPy, scikit-learn, pandas, MLflow

Systems & Cloud: FastAPI, PostgreSQL, Redis, DynamoDB, AWS EC2, S3, IAM, EBS/EFS, Docker, GitHub Actions

Visualization: Jupyter, Three.js, Mapbox, React

Engineering Practices: CI/CD, API design, integration testing, automated testing, reproducible experimentation

Specializations: Algorithms, ML systems, cloud infrastructure, scientific computing

TEACHING AND LEADERSHIP

CS Help Center Leader & Teaching Assistant | *University of Manitoba*

Jan. 2025 – Apr. 2025

- Mentored **100+** students across first- through third-year computer science courses and graded **150+** assignments for Analysis of Algorithms (COMP 3170).

Coding Kickoff Facilitator | *University of Manitoba*

Feb. 2025 – Mar. 2025

- Facilitated workshops that taught **IDE setup** and **Git** to **50+** first-year university students.

Events Executive | *.devClub (University of Manitoba)*

Aug. 2023 – Jul. 2024

- Co-led workshops, hackathons, academic review sessions, and community events for the Computer Science Department at the Faculty of Science.